Example model statement Car Registration Authority

Introduction

The model statement, which is given in sections Scope and Description, has been taken from ISO TR9007 *Information processing systems — Concepts and terminology for the conceptual schema and the information base*, 1987. This contains several renditions of the model using different kinds or representations including Entity-Relationship, A NIAM-like approach, and a formulation based on predicate logic.

Scope

The scope of the model to be described has to do with the registration of cars and is limited to the scope of interest of the Registration Authority. The Registration Authority exists for the purpose of:

- Knowing who is or was the registered owner of a car at any time from construction to destruction of the car.
- To monitor certain laws, for example regarding fuel consumption of cars and their transfer of ownership.

Description

Manufacturers of cars

There are a number of manufacturers, each with one unique name. Manufacturers may start operation, with the permission of the Registration Authority (which permission cannot be withdrawn). No more than five manufacturers may be in operation at any time. A manufacturer may cease to operate provided he owns no cars, in which case permission to operate lapses.

Cars

A car is of a particular model and is given a serial number by its manufacturer that is unique among the cars made by that manufacturer. The manufacturer is registered as owner of the car as soon as practicable. At this time it is given one registration number, unique for all cars and for all time. The year of production is also recorded. During the month of January only, a car can be declared to have been produced in the previous year. Eventually a car is destroyed and the date of destruction is registered. The history of the car must be kept until the end of the second calendar year following its destruction.

Car models

A model of car has one universally unique name. Cars of each model are made by only one manufacturer. New models may be introduced without limit. All cars of one model are recorded as having the same fuel consumption.

Fuel consumption

Fuel consumption is the number of liters of hydrocarbon fuel per 100 kilometers, which will be between 4 and 25 liters. The fuel consumption averaged over all registered cars produced by a particular manufacturer in a particular year is required not to exceed a maximum value, which is the same for each manufacturer and may change from year to year. At the end of each January an appropriate message is sent by the Registration Authority to each manufacturer which has failed to meet this requirement.

Garages

There are a number of garages [1], each one with a unique name. New garages may start trading. Garages may own cars, but at any time the cars they own must have originated from no more than three manufacturers (which three is unimportant, and may vary with time). A garage cannot cease to trade as long as it owns cars.

Persons

There are a number of persons who can own one or more cars. Each person has one unique name. Only those persons are of interest who own, or have at some time owned, a car still known to the Registration Authority.

Car ownership

At any one time a car may be owned by either its manufacturer, or a trading garage, or a person or a group of persons. If a car is owned by a group of persons, each is regarded as an owner.

Transfer of ownership

Ownership of a car is transferred by registration of the actual transfer, including the date. A manufacturer can only transfer to garages, and cannot be a transferee. A garage can transfer only to people. After destruction of the car it cannot be transferred anymore. Earlier transfer though still can be recorded.

[1] Dealerships in American English.